

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Equivalent expressions	Medium

Question

Which of the following is equivalent to $(1 - p)(1 + p + p^2 + p^3 + p^4 + p^5 + p^6)$?

Answer

- A. $1 - p^8$
- B. $1 - p^7$
- C. $1 - p^6$
- D. $1 - p^5$

Correct Answer: B

Rationale

Choice B is correct. Multiplying $(1 - p)$ by each term of the polynomial within the second pair of parentheses gives $(1 - p)1 = 1 - p$; $(1 - p)p = p - p^2$; $(1 - p)p^2 = p^2 - p^3$; $(1 - p)p^3 = p^3 - p^4$; $(1 - p)p^4 = p^4 - p^5$; $(1 - p)p^5 = p^5 - p^6$; and $(1 - p)p^6 = p^6 - p^7$. Adding these seven expressions together and combining like terms gives $1 + (p - p) + (p^2 - p^2) + (p^3 - p^3) + (p^4 - p^4) + (p^5 - p^5) + (p^6 - p^6) - p^7$, which can be simplified to $1 - p^7$.

Choices A, C, and D are incorrect and may result from incorrectly identifying the highest power of p in the expressions or incorrectly combining like terms.